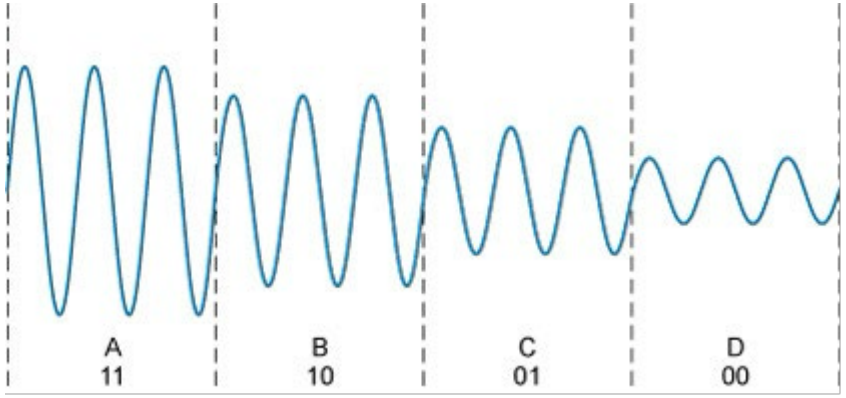
Tutorial 4

Q1

- What is the relationship between symbol rate and bit rate?
- The bits are the original digital data stream. Bit rate: the number of bits sent in one second, unit: bps
- In digital modulation, each symbol is typically a sine wave tone with a certain frequency, amplitude, and phase. Symbol rate is the number of transmitted symbols per second.
- One symbol can carry one or several bits of information.



$\boxed{N \cdot r \cdot S}$

- If N denotes the bit rate, S is the symbol rate, and r is the number of bits per symbol, then we have $N = r * S$. Also, $r = \log_2(L)$ where L the number of different symbols in total.

Q2

- Study the works of Shannon and Nyquist on channel capacity. Each places an upper limit on the bit rate of a channel based on two different approaches. How are the two related?
- Nyquist analyzed the theoretical capacity of a noiseless channel; therefore, in that case, the signaling rate is limited solely by channel bandwidth. Shannon addressed the question of what signaling rate can be achieved over a channel with a given bandwidth, a given signal power, and in the presence of noise.

Nyquist : Symbol rate (max) = 2B

因此最大 S 被 Bandwidth 限制

Q3

- How are binary values represented in amplitude shift keying, and what is the limitation of this approach?
- With amplitude-shift keying, binary values are represented by two different amplitudes of carrier frequencies. This approach is susceptible to sudden gain changes and is rather inefficient.

用移幅键控，二进制由两个不同的振幅表示，效率很低，

Q4

- What is QAM?
- QAM takes advantage of the fact that it is possible to send two different signals simultaneously on the same carrier frequency, by using two copies of the carrier frequency, one shifted by 90° with respect to the other. For QAM, each carrier is ASK modulated.

Q5

- What does the sampling theorem tell us concerning the rate of sampling required for an analogue signal?
- The sampling rate must be higher than twice the highest signal frequency.

BPSK $L=S$ → 二进制

QPSK $L=2S$ → 正交

QPSK = , $L=4$ $n=2$ 正交相移键控 — —

Q6

By N_y = $B = 2 \times 125{,}000 \cdot \log_2 L$

$B = 2 \times 2 \times 125{,}000 = 500\,kbps$

- Consider a noiseless channel with a bandwidth of 125 KHz. Assume that QPSK is used. What is the theoretical maximum bit rate that we can send through this channel? What is the theoretical maximum symbol rate?

So $S = 500 \div 2 = 250\,kbps$

- Since a noiseless channel is assumed, we can use Nyquist law to get the theoretical maximum bit rate. According to Nyquist law, we know that the theoretical maximum bit rate for a certain spectral bandwidth B (in hertz) is $2B log_2 L$, where B is the channel bandwidth, L is the number of different symbols.
-
- For QPSK, we have that L = 4. Therefore, for a noiseless channel with a bandwidth of 125 KHz, the theoretical maximum data rate is $2 * 125 * 10^3 * \log_2(4) = 500$ Kbps. Also, according to the relationship between symbol rate and bit rate, we get that the maximum symbol rate is actually equal to $2B = 2 * 125 * 10^3 = 250$ K symbols per second.
-

Q7

$$C = B \cdot \log_2(1 + SNR)$$

$$C = 125{,}000 \cdot \log_2\left(1 + \frac{7}{1}\right)$$

$$= 125{,}000 \times 3 = 375 \text{ kbps}$$

- Consider a noisy channel with a bandwidth of 125 KHz. Assume that the average received power is 7 watts and the average AWGN noise power is 1 watt. What is the theoretical maximum bit rate for this channel?
- Since we are now considering a noisy channel, then Nyquist law doesn't apply here. We need to use Shannon capacity to calculate the theoretical maximum data rate. It is known that the theoretical maximum data rate for a noisy channel subject to AWGN noise is given by Shannon capacity: $C = B\log_2(1 + S/N)$, where C is the channel capacity in bps, B is the channel bandwidth in hertz, S is the average received signal power in watts, N is the average noise power (including interference if any). S/N is the SNR, expressed in linear power ratio, not in dB.
- Therefore, we have that C = 125 * 10^3 * $\log_2(1 + 7/1)$ = 375 Kbps.
- Please note that we calculated the *theoretical maximum bit rates* for a noiseless and a noisy channel under some given conditions. In other words, the calculated data rates are the upper bounds of the practical data rate in practice.

Q8

- An analog signal has a bit rate of 8000 bps and a symbol rate of 1000 symbols per second. How many bits are carried by each signal element (symbol)? How many different symbols do we need?

$r = 8$

- According to N = r * S, we know that r = N/S = 8 bits per symbol. Therefore, each symbol carriers 8 bits. The number of different symbols $L = 2^r = 2^8 = 256$. Therefore, we need to have 256 different symbols.

$L = 2^8 = 256$

Q9

$r=1$ $L=2$

$600{,}000 = N$

- Suppose for a binary digital modulation, e.g., BPSK, the symbol rate is 600 Kbaud, then what is the bit rate?
- For binary modulation schemes, the symbol rate is equal to bit rate, because L=2 and $r = \log_2(L) = 1$. For binary digital modulations, there are in total two different symbols and one symbol is to represent one bit. For example, for BPSK, we can use phase of 0 to represent bit 0 and use phase π to represent bit 1.

bit rate is 600,000

Q10

$L=8$
$r=3$

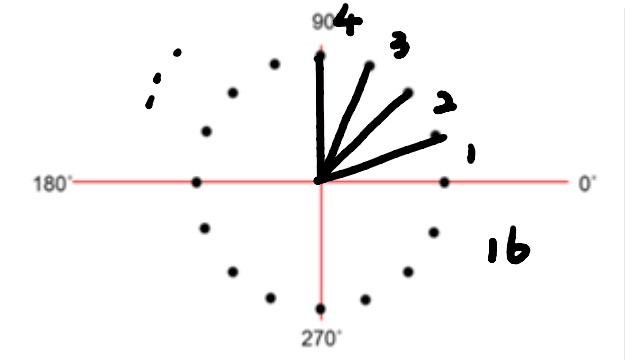
bit rate $= 3 \times 600$ kbps
$= 1800$ kbps

- If the baud rate (symbol rate) is 600 Kbaud for an 8PSK signal, then what is the bit rate?

- For 8PSK, L = 8 and r = 3 bits per symbol. There are 8 different symbols and each symbol represents 3 bits. Therefore, the bit rate N = r * S = 3 * 600 Kbaud = 1800 Kbps.
- Then, compared to BPSK in Q5, we can notice that with the same symbol rate, 8PSK transmits more bits.

Q11

1. Using the signal constellation diagram shown below, answer the following questions.
   a) What type of modulation does this represent?
   b) How many symbols are represented?
   c) How many bits per symbol are used? → 审题 bit per symbol
   d) If the Baud rate is 10,000 symbols/second, what is the bit rate?
   e) Would 16-QAM be more or less susceptible to noise than this type of modulation?



a = 16 PSK

b = 16 L = 16 r = 4

c = 4

d = 40,000

e = Yes, less

Ans

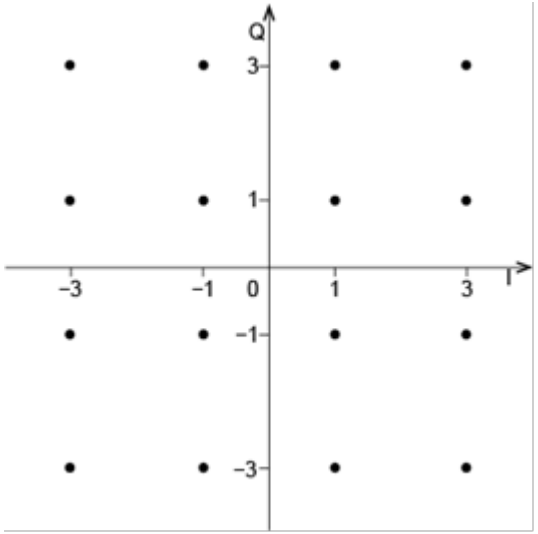
a) This constellation diagram represents 16-PSK, because there are in total 16 symbols in the constellation. Also, it shows that all the symbols appear on a circle about the origin. This is because for PSK, all the symbols have the same carrier frequency (not shown in the constellation) and amplitude (shown as the distance between the symbol and the origin in the constellation diagram).

b) 16 symbols are represented

c) Each symbol represents 4 bits.

d) The bit rate N = r * S = 4 * 10000 = 40000 bps.

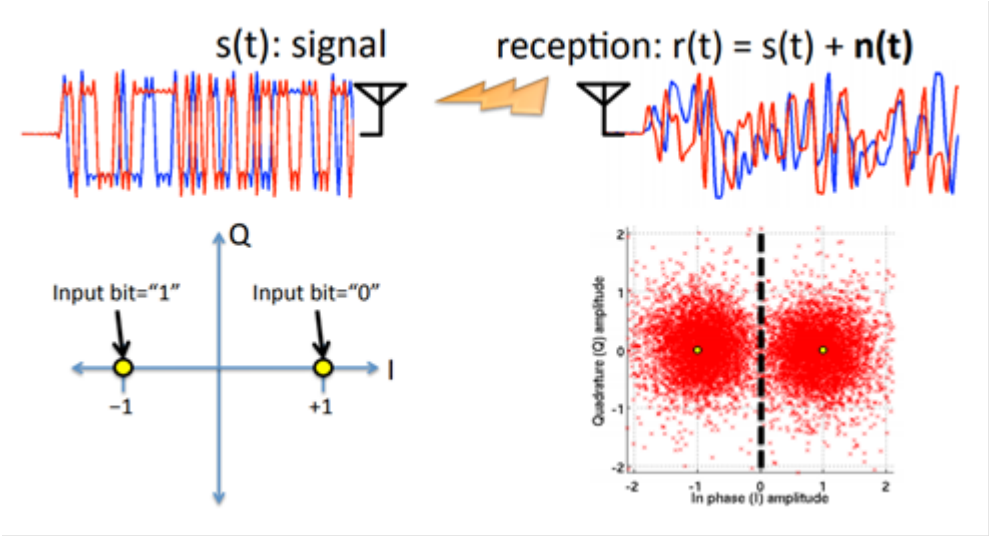
e) 16-QAM would be less susceptible to noise as the symbols are more spread out. Since QAM techniques modulate both the amplitude and the phase of the carrier so that the distance between symbols in a constellation can be increased, therefore QAM is more robust in the presence of noise. A constellation diagram for 16QAM is given below.



Cont'd

•

- The following figure shows what may happen with severe noise. It may be difficult for the receiver to determine the correct symbols that were transmitted. With the influence of severe noise, the receiver may mistakenly decide that it has received a bit 0 but actually it should be bit 1. When the symbols in the constellation diagram are closer to each other, it becomes more difficult for the receiver to distinguish them.



Q12

$C = 1M \cdot \log_2(1+63) = 6M$

$4M = 2 \cdot 1 \cdot \log_2(L)$, $L = 4$

Consider a channel with a 1 MHz capacity and an SNR of 63.

a. What is the upper limit to the data rate that the channel can carry?

b. The result of part (a) is the upper limit. However, as a practical matter, better error performance will be achieved at a lower data rate. Assume we choose a data rate of 2/3 as the maximum theoretical limit. How many signal levels are needed to achieve this data rate?

- **a.** Using Shannon's formula C = ~~106~~ $10^6 \times$ log2(1 + 63) = 6 MHz.
- **b.** Data rate = 4 MHz. Using Nyquist's formula 4 × 106 = 2 × 106 log2
- M
- M = ~~22~~ 2^2 = 4

Q13

- A channel can transmit signals in the frequency range of 2500 kHz to 4000 kHz. Compute its bandwidth. Assuming SNR_{dB} = 40, what will be the theoretical maximum capacity of this channel?
- Given SNRdB = 40, maximum frequency = 4000 KHz, minimum
- frequency = 2500 KHz
- Bandwidth, B = maximum frequency − minimum frequency
- = 4000 − 2500 = 1500 KHz
- SNRdB = 10 log10 SNR
- SNR = 10 SNRdB/10 = 1040/10 = 104 = 10,000
- By Shannon's Capacity, C = B log2(1+SNR)
- = 1500 log2 (1+10000)
- = 19,932 Kbps ≈ 20 Mbps

Q14

$$Att = 10 \cdot \lg \frac{P_{out}}{P_{in}}$$

- A 6-km-long cable has an attenuation of −0.25 dB/km. A signal with a power of 2.84 mW has been received at the end of the cable. Find out the power at which the signal was transmitted from the source.

Loss in decibels = 6 × −0.25 = −1.5 dB

$$10\log_{10}\frac{P_{out}}{P_{in}} = -1.5$$

$$\frac{P_{out}}{P_{in}} = 10^{-1.5/10}$$

P_{in} = 2.84 / 0.71 = 4 Mw

mw，不是 M

小的单位

Q15

- How many bits are needed for a QPSK signal? If the baud rate is 500 Mbaud, what will be the bit rate?
- Number of phases in QPSK = 4 $L=4$ $r=2$
- Hence, no. of bits transmitted per signal = log24 = 2
- Bit rate = Baud rate × no. of bits/signal = 2 × 500 = 1000 Mbps

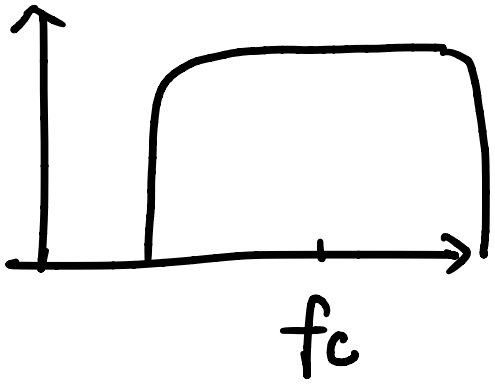
Q16

Given that d = 0.5, what is the required bandwidth for each of the following cases if the data rate is 8000 bps?

- a. ASK
- b. QPSK
- c. 64-QAM

ASK = r = 1, S = N = 8000

- a. For ASK, no. of bits per signal, n = 1
- Baud rate, S = R/n, where R is bit rate
- = 8000 baud
- Bandwidth, BT = (1 + d)S = (1 + 0.5) × 8000 = 12 kHz



- b. For QPSK, no. of signal levels, L = 4
- n = log2 4 = 2
- Baud rate, S = R/n , where R is bit rate
- = 8000/ 2 = 4000 baud
- Bandwidth, BT = (1 + d)S = (1 + 0.5) × 4000 = 6 kHz

- c. For 64-QAM, no. of signal levels, L = 64
- n = log264 = 6
- Baud rate, S = R/n , where R is bit rate
- = 8000/ 6 = 1333.33 baud
- Bandwidth, BT = (1 + d)S = (1 + 0.5) × 1333.33 = 2 kHz